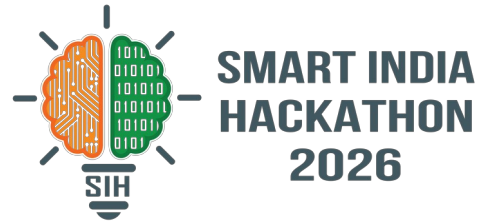


SMART INDIA HACKATHON 2026



AIRFARE APIx

PROBLEM STATEMENT

SIH26056

Development of a Real-time Airfare Price Index for India through Automated Web Scrapping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI)

Theme Smart Automation

Category Software

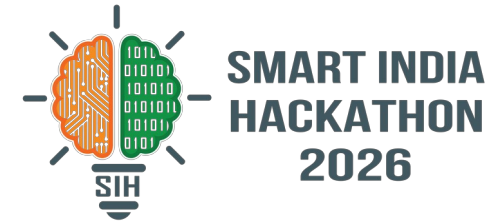
Team ID: _____

Team Name (Registered on portal): _____



TEAM
NAME

AIRFARE APIx



PROPOSED SOLUTION

A controlled daily airfare basket with every number traceable to raw evidence

Same routes, same fare definition and the same advance-booking windows. This makes daily, weekly and monthly movement statistically comparable.

ROUTES

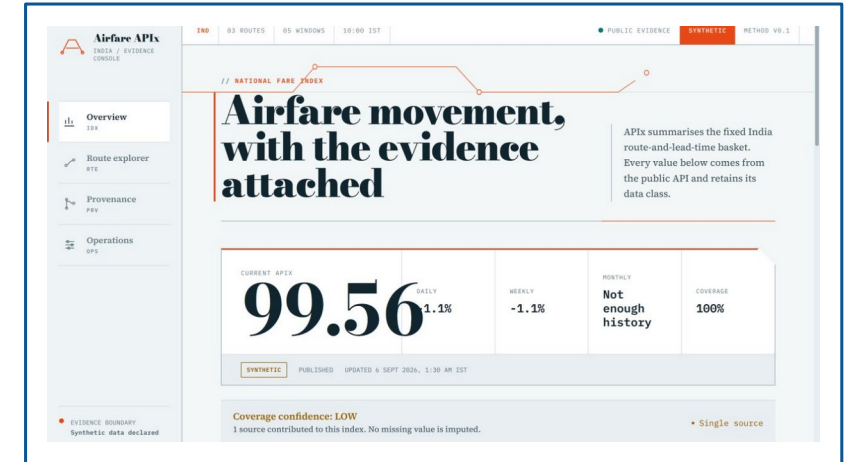
DEL-BOM DEL-BLR BOM-BLR

BOOKING WINDOWS

T+1 T+7 T+15 T+30 T+45

COMPARABLE FARE

1 adult One-way Economy Non-stop INR



WORKING PROTOTYPE / LIGHT UI

01 Evidence first

Raw quote + collection context remain attached.

02 Honest failures

Missing sources reduce coverage, never become zero.

03 One stored truth

Dashboard, API and exports use one result.



END-TO-END TRACEABILITY



INDEX METHOD

Current route-window median

Frozen base median

× 100

HONEST FAILURE PATH

Blocked source → job failure + source health

Missing basket cell → lower coverage + confidence

No failure is silently converted into a zero fare.

Python 3.12 • FastAPI • PostgreSQL • SQLAlchemy • APScheduler • React + Vite • Docker

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FEASIBILITY AND VIABILITY

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FEASIBILITY

Working locally today

Database, scheduler, collection pipeline, normalization, API, provenance, dashboard, exports and automated tests.

DEMO DATA

Deterministic synthetic fares keep the complete prototype testable without an external service.

PRODUCTION GATE

Live airline fares require approved API, GDS, aggregator or written web access.

CHALLENGE

Live fare access

MITIGATION

IndiGo/Air India NDC or approved partner feed

CAPTCHA and blocking

Record failure; never bypass or hide it

Different fare products

Validate cabin, currency, stop count and trip type

Sparse data

Publish coverage and confidence with APIx

Larger route basket

Add workers without changing data contracts

A transparent airfare measure for inflation analysis

01	NSO / MoSPI	Higher-frequency airfare evidence for CPI analysis with route and lead-time detail.
02	Researchers	Reproducible exports and raw-to-index provenance for independent checks.
03	Public and media	Clear trends that separate route effects from booking-window effects.

MEASURABLE OUTPUTS

5 booking windows

100 base index

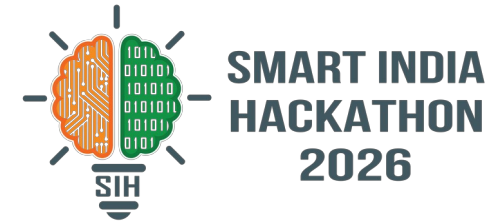
4 dashboard views

Scale next

DGCA-weighted routes and approved live sources.

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RESEARCH AND REFERENCES



01 SIH26056

Problem statement and expected solution

sih.gov.in

02 MoSPI eSankhyiki

Official CPI data and API documentation

esankhyiki.mospi.gov.in

03 DGCA

Domestic traffic reports for route evidence

dgca.gov.in

04 IndiGo NDC

Official AirShopping developer documentation

developer.goindigo.in

05 Air India NDC

Official direct-connect and partner routes

ndc.airindia.com

06 Duffel

Flights API test-mode limitations

duffel.com/docs

DEMO PATH

Overview → Route Explorer → Provenance → Operations

Show the index, inspect a route, trace one quote, then verify collection health.